

# P02 transport/workflow requalification — 2026-09-23

**Status: scoped local fixes verified; Block 2 remains in progress.** This is a local candidate receipt, not acceptance of release Stack 02, its predecessor, the assembled product, or a completed Copilot checkpoint.

## 1. Identity and frozen scope

- Owner: Codex /root, keepkey-vault-v11-agent-1; canonical coordination claim docs/release/audit-claims/715-02/ OWNER.md in the main firmware worktree.
- Main firmware worktree: /Users/highlander/WebstormProjects/keepkey-stack/projects/keepkey-firmware. Its checkout has unresolved conflicts and was not used for code editing.
- Main template SHA256 at intake after the claim:  
fbd340efde3c2e8ec0c0932c17e10b8c3b1592f3f25eb82562467d3ac7ae6d61.
- Isolated checkout: /private/tmp/kk715-p02-requalification-20260923; branch audit/p02-requalification-20260923.
- Frozen code/test candidate: 3ade6134877314293749af4e73de9ebf9d9de0c4. This report is generated after that commit. Later report-only commits do not change the checked implementation, tests, pins or workflows.
- Candidate base: 51411c4a7c96137bc80f8b99fa1690c4e385f7fd, published release Stack 02 [PR #832](#), branch release/715-stack-02-chains.
- PR #832's reported adjacent base at intake: 66ca726662822a9ff102bcd1354cc6a39d5a203b; 32 changed paths. This code-bearing PR is not synonymous with historical P02.
- Stack 01 [PR #831](#) current branch head: cec3cac2487356e0ffa556d201e5eab14f31e054. Stack 00b [PR #845](#) head: 4c56e19e3070772a28b2890a4ec862ecf77d5429. Both remained unaccepted at intake. Recheck before integration.
- Historical source: 614425a2a14d0113de251e7944118e47f31f5265. Historical receipt [PR #850](#) is documentation-only and cannot review this firmware delta.
- The Stack 02 source declares firmware **7.14.3**, despite its place in the 7.15 release program. Final host validation used FW\_VERSION=7.14.3 and the emulator reports that version. Release naming is not a capability assertion.
- Scope: the five P02 contracts below, their local regressions, and device receive paths. Exclusions: full chains-feature acceptance, unrelated P03/P04/EVM policy repairs, predecessor policy redesign, shared-branch restacking, release publication and physical-device qualification.
- No shared release branch was pushed, reset, retargeted or merged. No external review was requested. The candidate remains local pending accepted predecessor reconciliation.

### Dependency pins

No dependency pointer changed in this candidate. Required build dependencies were initialized at these exact pins; the combined old Stack 06 remediation's unpublished host pin was not imported.

| Dependency                     | Commit                                   |
|--------------------------------|------------------------------------------|
| code-signing-keys              | a6470bd8598e5e9a7bfc38bf139a5e5a616f05ec |
| deps/crypto/trezor-firmware    | 8a392f70a5d5575ece3dfb35f115d4a4b27f497c |
| deps/device-protocol           | dd9c85dc747cf965fb0e7bf49615dc9e7568ee65 |
| deps/googletest                | 7888184f28509dba839e3683409443e0b5bb8948 |
| deps/python-keepkey            | 49d537ce953b7524599bfc87f56b7a50a51a13a1 |
| deps/qrenc/QR-Code-generator   | 6dfbfdad5d9303ed190d1c3cb7bec34b565b6ce8 |
| deps/sca-hardening/SecAESSTM32 | 71d356a1141624994cf613bd2d2583892e8e6d5a |

Nested host device-protocol: dd9c85dc747cf965fb0e7bf49615dc9e7568ee65; token list: 89a64f717e1690bb31adb3e4c38e23640357333c. Unused nested crypto vendor/test repositories were not build inputs.

## 2. Behavior-to-adjacent-diff coverage

| Contract                                                                         | Published Stack 02 reconciliation                                                                                                                                                                                                                                 | Local candidate and coverage                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tiny credential bytes are erased and never reused                                | Retained from Stack 01: <code>messages.c</code> , <code>confirm_sm.c</code> , <code>pin_sm.c</code> , <code>passphrase_sm.c</code> , and USBRX. <code>TinyAcknowledgementDoesNotReusePreviousSecret</code> . Most wiping code is inherited, not new PR #832 code. | Existing wipe paths retained; native regression passes in both variants. Handler changes below preserve exit wipes. A future review must explicitly cover inherited wiping code.                                                                                                                                                                                           |
| Initialize/protected Ping revoke old signers; soft Initialize retains PIN policy | Initialize abort is already present in the Stack 00b base and retained. Protected Ping's explicit pre-wait abort is missing. Its regression was guarded.                                                                                                          | Add pre-wait signing abort without clearing PIN policy; ungate the Ping regression. New soft-Initialize test proves Bitcoin signing and passphrase are cleared while cached PIN remains, in both variants. The existing full-only multi-chain Initialize test also passes.                                                                                                 |
| Malformed/unexpected tiny input replies once and unwinds                         | Stack 01's rejected-tiny latch, loop exit and wipes are retained. Published Stack 06 regressions therefore do not justify replaying 189db1d2b wholesale. Valid foreign tiny acknowledgements and short packets still leave waits suspended.                       | Add waiting-handler type rejection and short-packet termination. Native tests and actual host replies cover malformed size, wrong-handler acknowledgements, normal requests in tiny mode, and short packets.                                                                                                                                                               |
| Synchronous receive packets are erased after callbacks                           | UDP, main USB, U2F and debug USB source retains the wipes. Stack 02 adds test receive injection; it does not supply all inherited packet hygiene.                                                                                                                 | USBRX. <code>PacketStorageIsWipedAfterCallback</code> executes in both ordinary variants. Main/debug short-transfer paths still wipe their full storage. U2F remains independently framed. Physical USB remains unverified.                                                                                                                                                |
| Dispatch boundaries revoke other signers and clear key scratch                   | Stack 02 defines <code>keepkey_before_message_dispatch</code> and <code>keepkey_after_message_dispatch</code> but <code>messages.c</code> never calls them. Three relevant regressions are guarded.                                                               | Restore normal dispatch hook calls, including cleanup; retain weak defaults for board-only linkage. Restore RAW dispatch hooks for parity; no active RAW entry was found in this candidate's firmware message map. Ungate the three regressions. All execute and pass in full; cross-chain acknowledgement coverage is excluded by the Bitcoin-only product configuration. |

### Historical Git inventory

Each table uses `git show --format= --numstat --no-renames COMMIT_SHA`. Counts include documentation, tests and gitlink lines where shown. These commits overlap; their totals are not the net candidate diff.

#### 71ee8c0bc949fb64f095cb42c930a8b462f3eb3d

Tiny credential copies previously survived consumption; wipe receive storage and handler copies and test stale-secret reuse.

```
Added Deleted Path
27      0 docs/release/audit-units/P02-001-tiny-lifetime.md
1       0 lib/board/confirm_sm.c
10      3 lib/board/messages.c
1       0 lib/firmware/passphrase_sm.c
1       0 lib/firmware/pin_sm.c
64      0 unittests/firmware/usb_rx.cpp
```

#### f20c2497a0990a6690c5bb804414c11cac74bf58

Initialize previously left signing workflows alive; abort workflows while preserving the intended cached-PIN policy.

```
Added Deleted Path
21      0 docs/release/audit-units/P02-002-initialize-abort.md
0       13 lib/firmware/fsm.c
2       6 lib/firmware/fsm_msg_common.h
17      2 unittests/firmware/fsm.cpp
```

#### 041a23d5c7fa7a09492644c4d55ab7f0c5ea22ad

Synchronous UDP and USB packets previously retained consumed data; wipe packet storage, including short-transfer exits.

```
Added Deleted Path
23      0 docs/release/audit-units/P02-003-packet-lifetime.md
3       0 lib/board/udp.c
```

|    |   |                               |
|----|---|-------------------------------|
| 14 | 3 | lib/board/usb.c               |
| 39 | 0 | unittests/firmware/usb_rx.cpp |

#### 5b2a62ce0e653a2478f7c4a74d3243535a02d576

Protected Ping response arena could contain DebugLink data; initialize its response after confirmation. Includes a historical host gitlink change, not replayed here.

| Added | Deleted | Path                          |
|-------|---------|-------------------------------|
| 1     | 1       | deps/python-keepkey           |
| 2     | 0       | lib/firmware/fsm_msg_common.h |

#### 32c6555316a203cb6c0b75a6553aafe56825089a

Rejected tiny input could leave its handler waiting or emitting a second response; latch rejection and unwind once.

| Added | Deleted | Path                             |
|-------|---------|----------------------------------|
| 3     | 0       | include/keepkey/board/messages.h |
| 24    | 5       | lib/board/messages.c             |
| 1     | 0       | lib/board/usb.c                  |

#### cb56504906bfe0470e3ce6601a6a77ad8e9dd309

Format the tiny-failure calls; no independent behavior change.

| Added | Deleted | Path                 |
|-------|---------|----------------------|
| 6     | 4       | lib/board/messages.c |

### Candidate changes and net code/test inventory

The baseline reproduction ran its original implementation with KK\_FINAL\_POLICY\_TESTS enabled and failed four selected tests. Applied source from ba52d68b6 supplies only protected-Ping cancellation; selected ae4061a6f hunks supply dispatch hooks and the three P02 test ungates. Its recovery/Ethereum/EOS progress and unlimited-approval changes were not copied. 189db1d2b supplies the malformed-size regression only, because its production tiny-loop fixes already exist in Stack 02.

- Dispatch/Ping fix 5e20b18c1: 19 dispatcher lines and 10 Ping lines restore the authorization boundaries; four regressions now execute in ordinary builds; add the malformed tiny-buffer regression.
- Foreign acknowledgement fix 7388977fe: PIN, passphrase and button handlers reject valid but inappropriate replies using the existing terminal latch, then unwind and erase their temporary copies. Debug state polling remains serviced. Three native negative-control tests distinguish the old behavior.
- Soft Initialize coverage a6827b7d8: explicitly verifies retained PIN, revoked Bitcoin signing and cleared passphrase in both configurations.
- Header-order correction 0cf8a82b4: C++ standard headers precede the firmware isprint macro in the new test includes.
- Host coverage 87447127e: validate one terminal wire response and an immediately usable Initialize round trip for each wrong acknowledgement and a normal request in tiny mode.
- Short-packet fix 2b1143768: reject short tiny packets in UDP and main/debug USB waits. Normal USB short-packet handling and U2F framing are preserved; no-data reads do not create a failure.
- Host short-packet test 3ade61348: checks raw UDP truncation gets exactly one Failure and unwinds to Initialize.

Net implementation/test range: `git diff --numstat`

51411c4a7c96137bc80f8b99fa1690c4e385f7fd...3ade6134877314293749af4e73de9ebf9d9de0c4. This is the direct P02 remediation delta, not PR #832's entire chains diff.

| Added | Deleted | Path                                      |
|-------|---------|-------------------------------------------|
| 4     | 0       | include/keepkey/board/messages.h          |
| 5     | 1       | lib/board/confirm_sm.c                    |
| 35    | 1       | lib/board/messages.c                      |
| 8     | 2       | lib/board/usb.c                           |
| 10    | 0       | lib/firmware/fsm_msg_common.h             |
| 8     | 0       | lib/firmware/passphrase_sm.c              |
| 3     | 0       | lib/firmware/pin_sm.c                     |
| 0     | 8       | unittests/firmware/fsm.cpp                |
| 21    | 0       | unittests/firmware/storage_passphrase.cpp |
| 110   | 1       | unittests/firmware/usb_rx.cpp             |
| 48    | 0       | unittests/host/test_p02_transport.py      |

### Complete local candidate inventory

Generation predecessor: 07e9389692eb4a708e31ebf662e59a0c4afac35b. Reproduce with `git diff --numstat`

51411c4a7c96137bc80f8b99fa1690c4e385f7fd...07e9389692eb4a708e31ebf662e59a0c4afac35b. This includes the report source,

binary PDF and artifact manifest. The final containing head is recorded in the canonical ledger; no remote review is claimed. Final committed path/count equality is checked after report regeneration.

| Added | Deleted | Path                                                                 |
|-------|---------|----------------------------------------------------------------------|
| 36    | 0       | docs/release/audit-units/P02-requalification-20260923-artifacts.json |
| 246   | 0       | docs/release/audit-units/P02-requalification-20260923.md             |
| -     | -       | docs/release/audit-units/P02-requalification-20260923.pdf            |
| 4     | 0       | include/keepkey/board/messages.h                                     |
| 5     | 1       | lib/board/confirm_sm.c                                               |
| 35    | 1       | lib/board/messages.c                                                 |
| 8     | 2       | lib/board/usb.c                                                      |
| 10    | 0       | lib/firmware/fsm_msg_common.h                                        |
| 8     | 0       | lib/firmware/passphrase_sm.c                                         |
| 3     | 0       | lib/firmware/pin_sm.c                                                |
| 0     | 8       | unittests/firmware/fsm.cpp                                           |
| 21    | 0       | unittests/firmware/storage_passphrase.cpp                            |
| 110   | 1       | unittests/firmware/usb_rx.cpp                                        |
| 48    | 0       | unittests/host/test_p02_transport.py                                 |

### 3. Findings and dispositions

| ID      | Origin and concrete evidence                                                                                                                                                                                                                     | Disposition                                                                                                                                                                                                                                                   |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P02-R01 | Missing dispatch calls on published Stack 02. Diagnostic baseline fails derived-key cleanup, cross-workflow acknowledgement, and top-level-confirmation signer cancellation.                                                                     | Fixed locally; all three regressions ungated and pass in their applicable ordinary variants.                                                                                                                                                                  |
| P02-R02 | Protected Ping enters a credential wait while old signing remains active; exact-baseline diagnostic fails the named test.                                                                                                                        | Fixed locally by aborting signing before protected Ping/authenticator waits. Ordinary shared regression passes.                                                                                                                                               |
| P02-R03 | Three waiting handlers silently ignore a valid acknowledgement for another handler. Negative control uses original handlers plus new tests, sends trailing Cancel to avoid a hang, and fails all three tests (no terminal rejection).            | Fixed locally; all three native cases and wire cases pass. No dependency change.                                                                                                                                                                              |
| P02-R04 | Tiny parser silently discards non-64-byte input; device main/debug receive callbacks discard short packets before that parser. Negative control leaves type at ERROR, failure count zero and caller buffer uncleared.                            | Fixed locally. Short UDP native/wire tests pass. ARM builds cover production main USB; debug USB was source-reviewed. Physical transfer testing remains required.                                                                                             |
| P02-I01 | Replay of Stack 02 51411c4a7 onto Stack 01 cec3cac24 conflicts in fsm.c, home_sm.c and unittests/firmware/fsm.cpp. Stack 01 uses sent-response progress; Stack 02 switches to explicit workflow progress.                                        | Pending predecessor integration. Replay was aborted after saving the conflict diff. Do not silently choose a progress policy or restack a shared branch.                                                                                                      |
| P02-I02 | Six remaining guarded policy tests were explicitly enabled on pre-short-packet candidate 87447127e. Feature polling passes; Bitcoin stream progress, recovery edits, Ethereum chunks, EOS data progress and padded-zero unlimited approval fail. | Five open inherited failures assigned to workflow-progress/P03/P04/EVM reconciliation. Their implementations and guards are unchanged by the final short-packet fix. They block assembled-candidate acceptance; this receipt claims only the P02 local scope. |
| P02-I03 | Repeating native xunit with the prior run's emulator.img fails Recovery.SpacesOnlyCeremonyIsRefusedAndCommitsNothing at its initial storage_isInitialized() assertion after storage_wipe(), in both variants.                                    | Pending storage/recovery isolation investigation. Logs and both prior images retained. Fresh-image ordinary suites pass, matching the initial isolated-container environment. This is not evidence of warm-image storage compatibility.                       |
| F161    | Historical candidate omitted the UDP packet regression from Bitcoin-only.                                                                                                                                                                        | Closed for this local candidate: usb_rx.cpp is unconditional and the named packet-wipe test executes in both variants.                                                                                                                                        |
| F204    | Shared UDP ports may cause interference.                                                                                                                                                                                                         | Controlled for these runs by one owned emulator/test process per isolated Docker network namespace with no published ports. Broader CI concurrency still requires its own evidence.                                                                           |

## 4. Verification and artifact identity

Builder: kktech/firmware@sha256:7438e53933d47d53157ed6d96d864cb208597e62dce26235ace09d1063427fa2, Linux amd64 under Docker on this ARM host. Final native and ARM source/test head is 3ade6134877314293749af4e73de9ebf9d9de0c4. Host dependency image: sha256:d1e9562baca19b277ec84fb45814ed4a32e9c08a11b95b86265eb450272db402. Host dependencies come from the pinned source checkout mounted into this immutable host-dependency image; firmware binaries are the audited build outputs, not that image's embedded firmware.

| Check                                     | Result                                 | Command / evidence                                                                                                                                                        |
|-------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Original baseline diagnostic              | 9 tests: 5 pass, 4 fail                | Original Stack 02 build with <code>-DKK_FINAL_POLICY_TESTS</code> ; selected P02 and USBRX tests; <code>baseline-focused.xml</code> .                                     |
| Foreign-ack negative control              | 3 expected failures                    | New tests with original handler/transport files overlaid; <code>foreign-baseline.xml</code> .                                                                             |
| Short-packet negative control             | 1 expected failure                     | New test with original receive implementation overlaid; <code>short-baseline.xml</code> .                                                                                 |
| Final full ordinary native                | 419 firmware, 19 board, 18 crypto pass | Emulator Debug clang/clang++ build; make xunit on fresh emulated flash.                                                                                                   |
| Final Bitcoin-only ordinary native        | 130 firmware, 19 board, 18 crypto pass | Same plus <code>-DKK_BITCOIN_ONLY=ON</code> ; make xunit on fresh emulated flash.                                                                                         |
| Final host protocol checks                | 13 pass, zero skipped per variant      | Ping, ClearSession and unittests/host/test_p02_transport.py; pinned host tests with <code>KK_FORCE_UDP=1</code> , actual <code>FW_VERSION=7.14.3</code> ; owned emulator. |
| Report validator                          | 6 pass                                 | Pinned host <code>test_report_variant_validation.py</code> .                                                                                                              |
| ARM full and Bitcoin-only                 | Both compile                           | Device cache, <code>-DCMAKE_BUILD_TYPE=MinSizeRel</code> ; production main USB compiled.                                                                                  |
| SRAM full                                 | PASS                                   | Repository tools/check_sram_budget.py; budget log below.                                                                                                                  |
| SRAM Bitcoin-only                         | PASS                                   | Same gate with <code>--variant bitcoin-only</code> .                                                                                                                      |
| Remaining guarded policy diagnostic       | 1 pass, 5 fail                         | Explicit diagnostic build; P02-I02 above.                                                                                                                                 |
| Repeat native with existing flash         | FAIL in both variants                  | Preserved as P02-I03, not counted as a passing compatibility check.                                                                                                       |
| Physical USB/OLED / signed-device upgrade | Not executed                           | Requires actual device qualification. Emulator tests cannot satisfy it.                                                                                                   |
| Exact-head hosted CI                      | Pending                                | No candidate was published or combined matrix dispatched while predecessor/policy gates remain open.                                                                      |

Artifact root: /private/tmp/kk715-p02-audit-evidence. The evidence manifest records SHA256 for logs, XML, emulator/native binaries and ARM ELF files. Build directories and prior flash images remain local. Host transport runner is `host-run.sh`; negative controls use source overlays under `foreign-baseline/`. Original diagnostic build initially lacked the nested token checkout and its first test launcher used unsupported BusyBox timeout syntax; both were corrected before the delivered baseline result recorded here.

### Artifact SHA256 manifest

| Artifact             | SHA256                                                           |
|----------------------|------------------------------------------------------------------|
| baseline-focused.log | 503d52f16f84e44640a60113a51e7497c255680dfbcd4a8aac8a8da9f6287472 |
| baseline-focused.xml | c3b30bf77360d3273d5c88911b16b3cecc1b63f7c98318cfc6d2673aa99ca471 |
| foreign-baseline.log | aade8ba542576aebf9ecbdc9070c2e009e566a256c9eb082eed17f7db4d2145  |
| foreign-baseline.xml | 5620b68e00840df826e8f5addfc091ccc8cda1a08a31645f7749d1bb2068a568 |

| Artifact                           | SHA256                                                            |
|------------------------------------|-------------------------------------------------------------------|
| short-baseline.log                 | 382b18ec4a2d9021b2560e1d1395819f69fedf0b273a6ff126ade4fba768d29e  |
| short-baseline.xml                 | 562c0fc4f770013bc04f72b8cabb9c28e394a277d93414371331fede1e165dc6  |
| full-xunit.log                     | 4ccbbf3b762d36f97d48f5d874d705b86685b5ba0ca9497f53c214d3f8a37688  |
| btc-xunit.log                      | 1b5c17d62097c8edb00f322bf749a9aa67771f0fabe63653fcebdc6cc93bac5c1 |
| full-xunit-existing-flash.log      | ff1f3c660e4175e7fa7e34a00c8ae9a27e7f067675a46b44b75a9e52943383    |
| btc-xunit-existing-flash.log       | 0700e90fe3f22656ddc09f8904228bc239fa923d6ca1a54250f1efeed2d745fe  |
| host-full.log                      | aac6b1fbee8eed669692a732699e933b5999aed0a8654daa9ee5e6b87db46475  |
| host-full.xml                      | b611fb84ec43b04aeba26cd084cfbb2bbaf64d7076a881a8bda82ea24660773c  |
| host-btc.log                       | c75f805dadf89b5e245e727c7610a25ef14bc2480c4c0f962261e78d34862c41  |
| host-btc.xml                       | 4698f40e4b906487d4288ce52d8c1bbe915fd4c7f7389d8b71b844f3a7c9ca60  |
| diagnostic.log                     | db5beaa21acdea7928b7eb9b253e9950ac5e57e5fdb5f6614ef59f607cb31135  |
| diagnostic.xml                     | 561568dd37cb2ed5e86b4e614b2de910e416d9212ed007cdda84413489fe1a58  |
| report-validation.log              | f9e6e463dd6852f16d4939afd481473269bde3bb6f570e01a31f0242ea18d1f1  |
| report-validation.xml              | d6fa8eb985a6e46a22f670e37dd5975efe31ccd95ae1b436bf6a2d23a6a9872c  |
| arm-full-budget.log                | 62236fa05c7e82d22b1577c2d724eafdd3c7a3874ad81f2eb9771200b8f94152  |
| arm-btc-budget.log                 | 88cd241ea83b7e26cc5a75ceee01d273bbe2f94b28f90d1c50eb9d2839988ef8  |
| predecessor-replay-conflicts.patch | 18bf2cb4b96a7bb74a4bf54fa2fb5d0d14bbefcfedf0162c43818675a17b12d5  |
| code-diff-gate.log                 | ee830a16b33bf8dc714b8183a626402c878824bca58f2efab2614fcf350e4a90  |
| host-run.sh                        | 829f0327027d71f3c53a3f123bcbb851d09a3ea9de7e2016f4ffff716283373e  |
| build-full/bin/firmware-unit       | b28e42082a43aa231134025b44883d135a860f8c6b58e8fa913d72739cddca88  |
| build-btc/bin/firmware-unit        | 7115755acb92e3297bb08043ea706b47a9aa03f14acce2346009739edde260d   |
| build-full/bin/kkemu               | ae9432a928085c7d6626ea8da41a5a045bfc3c0581ce67b95a16f1cc18c793e0  |
| build-btc/bin/kkemu                | ca3beac602fd9d001dda738397d0739d27988ce5192bcd1b332f8e9fa94f68ef  |
| arm-full/bin/firmware.keepkey.elf  | 626c8727e2915af71e3420a8fd082affff69221efd58994a8326a61b80dbe9ce  |
| arm-btc/bin/firmware.keepkey.elf   | 8aa316e79e2e8ab598a5552e9f721f2887110628059f27cb7158136bc9954c64  |

## 5. Report render and local preflight

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- The seven-page PDF was rendered from this Markdown and all pages were visually inspected. Tables, inventories and hashes are legible, with no clipped content.
- `git diff --check` passes on the frozen code/test range. The code-review machine gate identifies eligible implementation and test paths; it is not proof of inherited behavior coverage.
- No submodule pointer or workflow changes are included. Named P02 regressions execute in ordinary builds; remaining guarded tests and their failures are disclosed above.
- The published Stack 02 PR does not yet include this candidate. External-review preflight, final PR inventory including reports, live base/head alignment and accepted predecessor integration remain pending.
- Preserve separate status for local behavior verification, report review, code-review checkpoint, assembled product and release acceptance.

## 6. Review checkpoint and next integration action

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Copilot requests made in this audit: **zero**. Historical receipt reviews do not cover these changes. The SOP allows the late external checkpoint only after the release/upstream-unit acceptance conditions and owner authorization apply; no authorization was inferred from Stack 01's separate recorded request.

Next: resolve the Stack 00b/01 acceptance and workflow-progress interaction on the intended release sequence; port these bounded P02 fixes and regressions; retain or explicitly disposition all five contracts; revalidate the reconciled candidate. Close or assign the five inherited diagnostic failures and investigate the warm-image storage precondition before claiming the assembled candidate is clean. Prepare the final code-bearing adjacent report and PDF, verify its live file list, and request the authorized late review only after the SOP gate is satisfied.

Physical-device, signed-upgrade and release-promotion gates remain separate. Block 2 is not complete.